Mutriku wave power

Unpacking the full life-cycle footprint of breakwater wave energy

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Mutriku integrates oscillating water columns into a harbour breakwater in northern Spain. It combines electricity generation with turbine testing. This case asks whether improving output can justify the material, ecological and societal burdens of building, operating and eventually retiring the plant [1,4].

How oscillating water column technology evolved

Sources: [3,7,8]

1910 — Bochaux Praceique builds the believed first oscillating water column (OWC)

1940s — Masuda develops a wave-powered navigation buoy

Late 1970s — Wells turbine invented: airflow both ways, no rectifying valves needed

1985 — Toftestallen, Norway: full-size shoreline OWC prototype

1999–2000 — Pico (Azores) and LIMPET (Scotland) plants; LIMPET ran over 10 years

2011 — **Grid connection: 16 × 18.5 kW turbines (296 kW) [1,14]**

Mutriku integrates the OWC into a breakwater, so one structure also protects the harbour.

Did you know?

Europe's first: Mutriku was the first commercial plant in Europe to generate electricity from waves [13].

Longest-running: since 2011 it has logged more operating hours than any other wave plant of its kind [13].

Open to visitors: the plant is part of the tourist offer of the UNESCO-recognised Basque Coast Geopark [14].

01 How it works



1. Wave lifts the water column
2. Air is pushed through the turbine
3. Wells turbine keeps turning one way
4. Generator makes electricity [1]

16 × 18.5 kW = 296 kW [1,14]

Simplified section, not to scale. Arrows show alternating motion. Diagram based on [1].

02 Where it sits: location and settings



Photo: Txo (2026), Wikimedia Commons, CC0 1.0

Location: inside the harbour breakwater at Mutriku, on the Basque coast of northern Spain.

Why there: built into the breakwater, one structure shares harbour protection, port use and electricity generation [1,3]. The BiMEP Mutriku area also hosts turbine testing [4,9].

03 Where the carbon burden sits

226 g CO_2e/kWh — Life-cycle total, 20-year model [1]

? Dismantling emissions not quantified (set to 0)



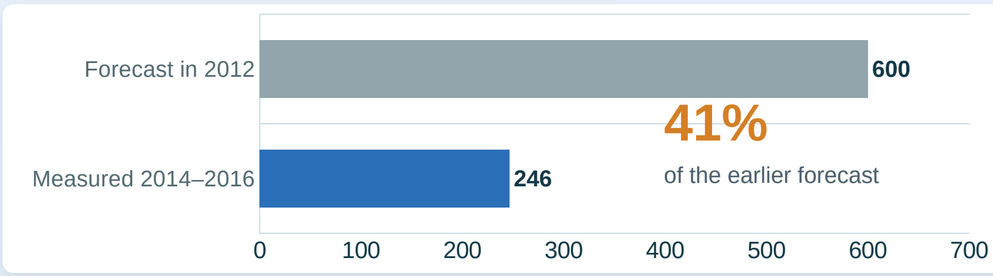
- **Construction 86.6%**
- **Operation 5.5%**
- **End of life 7.9%**

Secondary inventories | Staff data | Dismantling set to 0 (no data)

Shares use rounded stages: 195, 12.4 and 17.9 g CO_2e/kWh. Their sum is 225.3, versus 226 reported [1]. Model: 20 years and 4.93 GWh [1].

Construction dominates, so material demand is the priority. The model includes breakwater construction but excludes its maintenance [1].

04 Output: expectation and evidence



Annual electricity, MWh/year. Historical forecast versus measured mean [2,3].

About 10 of 16 turbines working on average [2]

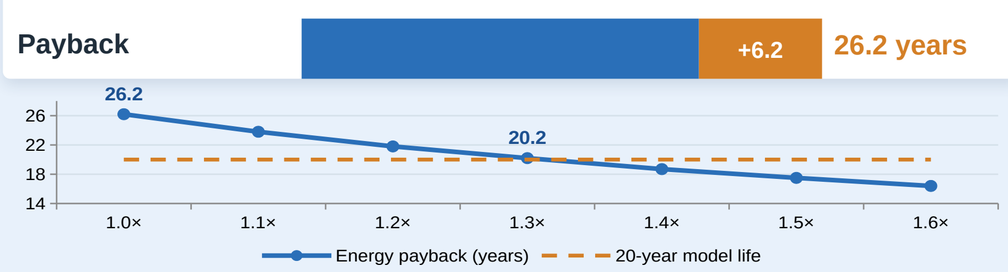
Measured output was about 41% of the earlier forecast [2,3]. Cumulative grid generation passed 3 GWh by the end of 2023 [13]. Different conditions prevent isolating turbine design, downtime or wave climate.

Beyond carbon: the wider footprint

▲ Favourable ▼ Adverse – Neutral

- ▼ Storm-time construction noise was reportedly heard 3–10 km away; locals were concerned [10]
- – Breakwater may alter sediment and flow; the OWC adds no significant change [10]
- ▲ Air-driven turbines never touch seawater; sits on the breakwater, not natural seabed [11,12]
- ▼ Most other LCA categories (e.g. marine ecotoxicity) are also driven by construction [1]

05 Energy payback



Extend service life | Raise net output | Use less material

Chart x-axis: annual output relative to the model. Break-even needs about 31% more output, or a service life of about 26 years (26.2 ÷ 20). Author's scaling; assumes embodied energy is unchanged [1].

06 Impacts, trade-offs and mitigation

	Environment (▲ Favourable ▼ Adverse ? Not assessed)	Society (● Modelled ● Source-reported ● Unsourced or unresolved)
Build	▼ 195 g CO_2e/kWh; civil works disturb habitats [1] ●	▼ Works and traffic disturb residents and port users ●
Operate	▲ Low emissions: 12.4 g CO_2e/kWh [1] ● ▲ No clear sound increase at 1 km [5] ● ▼ Near-field sound, species effects unresolved [5] ●	▲ Electricity, research access, skills, harbour protection ● ▲ Visitor interest suggests a local benefit [3] ● ▼ Public cost, worker exposure, unequal access ●
Retire	▼ Removal disturbs habitats; dismantling emissions modelled as zero [1] ● ▲ Reuse or recycling may avoid virgin production ●	? Not assessed here. Jobs, current acceptance unverified; 2013 study dated [6] ●
Fix	Cut virgin materials | Verify marine durability | Upgrade turbines [4] | Compare end-of-life options	Coordinate work schedules | Track output and complaints | Residents shape decisions [5,6]

Community milestones

Selected milestones, not a time-scaled axis.

- Jul 2011 – Plant opens [3]
- Dec 2011 – Storm shutdown [3]
- Mar 2012 – Operation resumes [3]
- 2013 – Social study [6]
- 2025 – Engagement report [5]

Selection prioritised recent primary studies on impacts, output and turbines. Historical accounts provide context. This focused review is not systematic.

Conclusions and Recommendations

Key findings

1. **Construction dominates:** 86.6% of the 226 g CO_2e/kWh life-cycle total [1].
2. **Payback exceeds service life:** 26.2 years against a 20-year model life; output was ~41% of forecast [1–3].
3. **Evidence is incomplete:** dismantling is modelled as zero [1]; social evidence is dated [6].

Key judgement

Wave electricity alone does not make Mutriku sustainable. It is promising, but only lower material use, higher output or longer life, with full life-cycle accounting and fair sharing of burdens, can justify it.

1. Upgrade within the existing structure
2. Compare equivalent alternatives
3. Test lifetime, downtime, recycling
4. Approve only with full accounting
5. Publish monitoring; revisit

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